Unit 5 · Session 2 · Thu, Nov 19, 2026

Renal Pharmacotherapeutics — Session 2

4 parts · every fact cited to its slide or textbook page. About 130 minutes of focused study.

Study plan

20 min
Read the guide
One pass, top to bottom. Bold = what the lecture emphasized.

65 min
Diagrams & tables
Redraw each flow from memory; cover a table column and recall it.

10 min
Cram sheet
Hide answers and quiz yourself. Repeat until it's automatic.

30 min
Self-test
Answer before you look. Read every explanation, even when right.

5 min
Objectives check
Tick each objective only if you can teach it out loud.

HOW TO READ THIS GUIDE

- **Bold** = emphasized on the lecture slides
- TB textbook only · SG course study guide only
- S12 = slide 12 · S12 (notes) = speaker notes · SG U1-R3 = study guide item · p. 17 = Katzung page
- Open **Sidekick** (bottom right) to search, listen, drill flashcards, quiz, or chat about this lecture

The whole lecture in 6 sentences

1. The **RAAS**: angiotensinogen → (renin) → angiotensin I → (ACE) → angiotensin II → vasoconstriction + aldosterone; aliskiren, ACE inhibitors, ARBs and aldosterone antagonists each block one step. S4; SG U5-R7

2. **ACE inhibitors/ARBs** are first-line in proteinuric CKD: efferent dilation lowers glomerular pressure; a creatinine rise **up to ~30%** is expected; check K^+/creatinine in 1–2 weeks; **never in pregnancy**. S5; S8; S10; SG U5-R8; SG U5-R9

3. ACE-inhibitor **cough** is bradykinin, not allergy → switch to an **ARB**; angioedema → stop. S6; S7

4. **Non-DHP CCBs + beta-blockers** → bradycardia/AV block; resistant HTN in CKD is often a thiazide that needs to become a **loop**. S13; S15; S17

5. **Nitrates**: NO → cGMP → venodilation; nitrate-free interval; **never with PDE5 inhibitors**. **CCBs** treat both hypertension and angina by the same mechanism. S19; S20; S22; SG U5-R10; SG U5-R11

6. **Triple whammy**: NSAID + ACEi/ARB + diuretic → AKI; counsel against OTC NSAIDs. S23; S27; SG U5-R12; SG U5-S

Learning objectives

IO = syllabus unit objective · SO = this session's objectives slide.

IO5.1 Identify major pharmacologic classes used in the management of renal conditions and describe their mechanisms of action.

Part 1 · RAAS Blockers
Part 2 · Other Antihypertensives & Resistant Hypertension
Part 3 · Nitrates, Angina & the Triple Whammy

IO5.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat renal conditions.

Part 1 · RAAS Blockers
Part 2 · Other Antihypertensives & Resistant Hypertension
Part 3 · Nitrates, Angina & the Triple Whammy

IO5.3 Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.

Part 2 · Other Antihypertensives & Resistant Hypertension
Part 3 · Nitrates, Angina & the Triple Whammy
Part 4 · Putting It All Together: Renal Toolkit

IO5.4 Select appropriate pharmacologic therapy for common renal conditions based on clinical presentation and patient-specific factors.

Part 1 · RAAS Blockers
Part 2 · Other Antihypertensives & Resistant Hypertension
Part 4 · Putting It All Together: Renal Toolkit

IO5.5 Identify appropriate dosing, administration, and monitoring considerations for medications used in renal conditions.

Part 1 · RAAS Blockers

IO5.6 Explain important medication-use and safety information to patients receiving pharmacotherapy for renal.

Part 3 · Nitrates, Angina & the Triple Whammy
Part 4 · Putting It All Together: Renal Toolkit

SO1 Describe the renin-angiotensin-aldosterone system (RAAS) and identify where each antihypertensive drug class acts.

Part 1 · RAAS Blockers

SO2 Explain why ACE inhibitors and ARBs are preferred first-line therapy for hypertension in chronic kidney disease, including their renoprotective mechanism.

Part 1 · RAAS Blockers

SO3 Identify major antihypertensive drug classes beyond RAAS blockers, their mechanisms, and their role in CKD-specific management.

Part 2 · Other Antihypertensives & Resistant Hypertension

SO4 Describe vasodilator pharmacology (nitrates, calcium channel blockers) in angina pectoris and its mechanistic overlap with cardiovascular-renal disease management.

Part 3 · Nitrates, Angina & the Triple Whammy

SO5 Identify high-risk drug combinations and monitoring needs at the cardiorenal interface, including the "triple whammy," and apply the full renal/cardiorenal toolkit to integrated clinical scenarios with a cumulative recall quiz.

Part 3 · Nitrates, Angina & the Triple Whammy
Part 4 · Putting It All Together: Renal Toolkit

ABOUT THIS LECTURE

- **PA PHARMACOLOGY COURSE · LECTURE 5 · SESSION 2 OF 2** — Renal Pharmacotherapeutics. **Session 2: Hypertension in CKD and the Cardiovascular-Renal Overlap | Thursday, November 19 | 8:30–11:00 AM** S1
- Recap: Session 1 covered renal dosing principles and the five diuretic classes, organized by nephron site — Session 1 gave us nephron sites — plus the hyperkalemia-with-ACEi/ARB preview from the closing case. S1 (notes); S2; S2 (notes)
- Today we follow one pathway, the renin-angiotensin-aldosterone system, then see where it intersects with angina drugs and with the most preventable cause of AKI, the triple whammy — closing with a cumulative recall quiz spanning both sessions. S2 (notes)
- Appendix 1 abbreviation key: PO = by mouth; IV = intravenous; SL = sublingual; BID / TID / QID = two / three / four times daily; ER / SR = extended-release / sustained-release; IR = immediate-release; HF = heart failure; MI = myocardial infarction. S36; S36 (notes)
- LV = left ventricular; EF = ejection fraction; AV = atrioventricular; SVT = supraventricular tachycardia; WPW = Wolff-Parkinson-White; CAD = coronary artery disease; ECG = electrocardiogram; ICP = intracranial pressure; CBC = complete blood count. S36; S36 (notes)
- ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; AKI = acute kidney injury; CKD = chronic kidney disease; CrCl = creatinine clearance; eGFR = estimated glomerular filtration rate; PDE5 = phosphodiesterase type 5; CYP = cytochrome P450. S36; S36 (notes); S55 (notes)
- The profiles give **Formulation, dose, indications, adverse effects, contraindications and toxicity** — your study reference: know the dose logic, the serious adverse effects, the contraindication that changes decisions and the toxicity you would need to recognize. **Standard adult dosing for teaching; verify against current prescribing information.** S36; S36 (notes); S37-54
- **Appendix 2: DRUG PROFILE · SUPPORTING** — supporting drugs explain interactions, not doses. **Listed for context; not profiled in detail.** S55; S55 (notes)
- **Original figures in this deck:** Figures 8, 12, 15, 16 and 18 (The Renin-Angiotensin-Aldosterone System: Where Antihypertensives Act; Antihypertensive Classes: BP-Lowering Efficacy vs. Renoprotection in CKD; Angina Pathophysiology: Where Nitrates and Calcium Channel Blockers Act; S56
- The "Triple Whammy": A Classic Preventable Cause of AKI; The Cardiorenal-Hypertension Cycle) were generated by the course director using the AI tool Claude to illustrate standard, non-copyrighted pharmacologic concepts — original diagrams, not reproduced from Katzung's Basic and Clinical Pharmacology or any other copyrighted source. S56; S56 (notes)
- **Mechanism illustrations:** Figures 9, 10, 11, 13, 14 and 17 are illustrative AI-generated schematics of drug mechanisms, not reproduced from any copyrighted source. **Reference:** Anthropic. (2026). Claude (Sonnet 5) [Large language model]. https://claude.ai S56; S56 (notes)

Slide discrepancies: none found (slides, speaker notes, and study guide agree).

1 PART 1 · DRUG CLASS

RAAS Blockers

SO1 SO2 IO5.1 IO5.2 IO5.4 IO5.5

CLASS

- **One pathway, four drug classes, and the preferred therapy for hypertension in CKD.** Today we follow the renin-angiotensin-aldosterone system, the pathway these drugs act on. S3; S3 (notes)
- **Figure 8. Original illustrative diagram (not from Katzung): angiotensinogen is converted to angiotensin I by renin, then to angiotensin II by ACE; angiotensin II drives both vasoconstriction and aldosterone release.** S4; SG U5-R7
- **Direct renin inhibitors, ACE inhibitors, ARBs, and aldosterone antagonists each block a different point in this cascade.** S4; SG U5-R7
- This is the conceptual spine for the session: walk the cascade before naming drugs — angiotensinogen, renin, angiotensin I, ACE, angiotensin II, vasoconstriction and aldosterone. S4 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- ACE inhibitors: captopril, lisinopril, enalapril — "-pril" suffix across the class. ARBs: losartan, valsartan — "-sartan" suffix. Direct renin inhibitor: aliskiren. S5; S7

MECHANISM OF ACTION

- **Mechanism: block the conversion of angiotensin I to angiotensin II, reducing both vasoconstriction and aldosterone release (Figure 8).** S5
- ACE also breaks down bradykinin: ACE inhibitor blocks ACE → less angiotensin II (less vasoconstriction and aldosterone) → less bradykinin breakdown, so bradykinin accumulates → cough or angioedema → switch to an ARB for cough. S6; S6 (notes)

INDICATIONS

- **ACE inhibitors and ARBs are first-line antihypertensive therapy in CKD, especially with proteinuria — not simply because they lower blood pressure.** S8
- ARBs: no bradykinin-related cough — preferred alternative when ACE inhibitor cough is intolerable. ARBs do not raise bradykinin, so they are the usual switch. S6; S7

- ARBs block the AT1 receptor directly, rather than preventing angiotensin II formation. Aliskiren blocks the rate-limiting first step of the entire cascade. S7
- Renoprotection: by reducing angiotensin II, these drugs preferentially dilate the efferent (outflow) arteriole of the glomerulus, lowering intraglomerular pressure and reducing proteinuria — a direct, structural renoprotective effect beyond systemic BP control. S8; SG U5-R8
- Chain (Figure 10): angiotensin II normally constricts the efferent arteriole → ACE inhibitor or ARB removes that signal → efferent arteriole dilates; intraglomerular pressure falls → proteinuria falls; creatinine rises up to ~30%. S9

- Aliskiren: limited clinical use; useful mainly as a teaching example of where a drug could act upstream of ACE — included for conceptual completeness on Figure 8. S7; S7 (notes)
- The drug class students might expect to avoid in kidney disease is actually the preferred therapy, precisely because of how it works at the glomerulus. S8 (notes)

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- **Adverse effects: dry cough (bradykinin accumulation — the single most common reason for switching to an ARB), angioedema (rare but potentially life-threatening), and hyperkalemia.** S5
- Cough is bradykinin-mediated, not allergic. Angioedema is rare but can be life-threatening: stop the drug. S6
- An initial rise in serum creatinine of up to ~30% after starting therapy is expected and acceptable — reflects reduced intraglomerular pressure, not drug-induced kidney injury; a larger rise warrants evaluation for renal artery stenosis. S5; SG U5-R9
- Students frequently get the creatinine point wrong: many want to stop the ACE inhibitor at the first rise. It is a predictable hemodynamic effect, not toxicity — the mechanism and the "side effect" are two faces of the same action. S5 (notes); S8; S9 (notes)

CONTRAINDICATIONS

- **Contraindicated in pregnancy — fetal renal toxicity; avoid in women who are or may become pregnant.** ARBs share the same pregnancy contraindication and hyperkalemia risk as ACE inhibitors. S5; S7
- Aliskiren: generally not combined with an ACE inhibitor or ARB due to additive hyperkalemia/hypotension risk with no added benefit — never in diabetes. S7; S7 (notes)

DRUG INTERACTIONS

- Potassium supplements, potassium-sparing diuretics and trimethoprim: hyperkalemia with ACE inhibitors and ARBs. NSAIDs blunt antihypertensives and raise AKI risk with an ACE inhibitor or ARB plus diuretic. S55; S55 (notes)
- TB Hyperkalemia with potassium-sparing diuretics rises with drugs that cut renin or angiotensin II: β blockers, NSAIDs, aliskiren, ACE inhibitors, ARBs. p. 275

PHARMACOKINETICS

- TB Enalapril's ACE-inhibition half-life is about 4 hours, yet once-daily dosing works: concentrations stay far above the C50, so ACE stays inhibited. p. 50

MONITORING

- **Monitor potassium and renal function within 1–2 weeks of starting or up-titrating therapy, and periodically thereafter.** S8; S9 (notes)
- **Hold or use with caution during acute illness causing volume depletion ("sick day rules") — combined with reduced renal perfusion, RAAS blockade can precipitate acute kidney injury.** S8
- Sick-day rules: hold the ACE inhibitor or ARB during vomiting, diarrhea or dehydration. An early creatinine rise is expected and hemodynamic, not nephrotoxicity; a larger rise warrants checking for volume depletion or renal artery stenosis. S9; SG U5-R9

CLINICAL PEARL / WHY

- Case: a patient with hypertension and diabetic nephropathy is started on lisinopril; two weeks later serum creatinine has risen from 1.1 to 1.4 mg/dL. **Is this an indication to stop the ACE inhibitor?** S10
- **A rise of roughly 27% is within the expected range for ACE inhibitor initiation, reflecting reduced intraglomerular pressure from efferent arteriolar dilation — not drug toxicity.** S10; S10 (notes)
- **Therapy should generally continue with ongoing monitoring; a rise beyond approximately 30%, or a much sharper increase, should prompt evaluation for renal artery stenosis or volume depletion.** S10
- A student who reflexively says "stop the drug" has missed that this creatinine rise is expected and even reflects the mechanism working as intended. S10 (notes)

Step by step

Creatinine rose after starting an ACE inhibitor S8; S9; S10

Recheck K^+ and creatinine at 1–2 weeks →

Rise ≤ ~30%? Expected — continue and monitor →

Rise > 30% or sharp? Look for volume depletion or renal artery stenosis →

Vomiting, diarrhea, dehydration → sick-day rules: hold it

Figure 8 · The RAAS: where antihypertensives act. Original illustrative diagram (not from Katzung): angiotensinogen → (renin) → angiotensin I → (ACE) → angiotensin II → vasoconstriction and aldosterone. Direct renin inhibitors, ACE inhibitors, ARBs and aldosterone antagonists each block a different step. S4; S4 (notes)

Figure 9 · ACE inhibitors: bradykinin, cough and angioedema. Schematic (illustrative) — Figure 9. ACE inhibition and bradykinin accumulation: ACE also breaks down bradykinin, so blocking it raises bradykinin (cough, angioedema); ARBs spare bradykinin. S6; S6 (notes)

Figure 10 · Efferent arteriole dilation: why creatinine rises. Schematic (illustrative) — Figure 10. Efferent arteriole dilation and intraglomerular pressure: removing angiotensin II's efferent constriction lowers glomerular pressure — less proteinuria, and an expected creatinine rise of up to ~30%. S9; S9 (notes)

ACE inhibitors vs. ARBs vs. aliskiren

	ACE inhibitors (-pril)	ARBs (-sartan)	Aliskiren (-kiren)
Block	Angiotensin I → II conversion	AT1 receptor	Renin (first step)
Cough / bradykinin	Yes — most common reason to switch	No	Cough listed on profile
Pregnancy	Contraindicated	Contraindicated	Contraindicated
Use	First-line HTN in proteinuric CKD	First-line; ACEi cough alternative	Limited; never with ACEi/ARB in diabetes

How to tell them apart. Sources: S5; S7; S37-42

ACE inhibitor profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Captopril (Capoten)	**ACE inhibitor**	Tablets 12.5/25/50/100 mg	Hypertension: 12.5-25 mg PO BID-TID (maximum 450 mg/day); heart failure: 6.25-12.5 mg TID	Hypertension, heart failure, post-MI, diabetic nephropathy
Lisinopril (Zestril, Prinivil)	**ACE inhibitor**	Tablets 2.5/5/10/20/30/40 mg	Hypertension: 10 mg PO daily, titrate to 40 mg (maximum 80 mg); heart failure: start 2.5-5 mg daily	Hypertension, heart failure, post-MI, diabetic nephropathy
Enalapril (Vasotec)	**ACE inhibitor (prodrug)**	Tablets 2.5/5/10/20 mg; IV enalaprilat	Hypertension: 5 mg PO daily, titrate to 40 mg daily or divided; heart failure: 2.5 mg BID	Hypertension, heart failure, asymptomatic LV dysfunction

How to tell them apart. Sources: S37; S38; S39

ACE inhibitor profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Captopril (Capoten)	Cough, hyperkalemia, hypotension, taste change, rash, angioedema	Pregnancy, history of angioedema with ACE inhibitors, bilateral renal artery stenosis, concomitant aliskiren in diabetes	Angioedema, hyperkalemia, hypotension, AKI; take 1 hour before meals; monitor potassium and creatinine
Lisinopril (Zestril, Prinivil)	Dry cough, dizziness, hyperkalemia, creatinine rise up to 30%, angioedema	Pregnancy, angioedema history, bilateral renal artery stenosis, concomitant aliskiren in diabetes	Angioedema, hyperkalemia, symptomatic hypotension, AKI during volume depletion
Enalapril (Vasotec)	Cough, hyperkalemia, dizziness, hypotension, angioedema	Pregnancy, angioedema history, bilateral renal artery stenosis	Hypotension, hyperkalemia, angioedema, AKI; reduce dose when CrCl is below 30 mL/min

How to tell them apart. Sources: S37; S38; S39

ARB and renin inhibitor profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Losartan (Cozaar)	**Angiotensin receptor blocker (ARB)**	Tablets 25/50/100 mg	Hypertension: 50 mg PO daily, up to 100 mg; diabetic nephropathy: 50 mg daily, increase to 100 mg	Hypertension, diabetic nephropathy, stroke risk reduction with LV hypertrophy, heart failure alternative
Valsartan (Diovan)	**Angiotensin receptor blocker (ARB)**	Tablets 40/80/160/320 mg	Hypertension: 80-160 mg PO daily (maximum 320 mg); heart failure: 40 mg BID, titrate to 160 mg BID	Hypertension, heart failure, post-MI
Aliskiren (Tekturna)	**Direct renin inhibitor**	Tablets 150/300 mg	150 mg PO daily, may increase to 300 mg daily; take consistently with respect to meals	Hypertension (rarely first-line)

How to tell them apart. Sources: S40; S41; S42

ARB and renin inhibitor profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Losartan (Cozaar)	Dizziness, hyperkalemia, creatinine rise, rare angioedema	Pregnancy, concomitant aliskiren in diabetes, bilateral renal artery stenosis	Hypotension, hyperkalemia, fetal toxicity; no bradykinin cough
Valsartan (Diovan)	Dizziness, hyperkalemia, fatigue, creatinine rise	Pregnancy, concomitant aliskiren in diabetes, bilateral renal artery stenosis	Hypotension, hyperkalemia, fetal toxicity
Aliskiren (Tekturna)	Diarrhea, cough, hyperkalemia, angioedema, hypotension	Pregnancy; do not combine with ACE inhibitors or ARBs in diabetes or with eGFR below 60 mL/min	Hypotension, hyperkalemia, AKI when combined with ACE inhibitors or ARBs

How to tell them apart. Sources: S40; S41; S42

WHY IT HAPPENS

- Why does creatinine rise on an ACE inhibitor? The drug lowers glomerular pressure on purpose — that same drop protects the kidney from proteinuria. S8; S9; SG U5-R8

MNEMONIC (MEMORY AID)

- Thirty is fine, more is a sign: up to ~30% creatinine rise → continue; more → look for renal artery stenosis or volume depletion. S5; S10; SG U5-R9

HIGH-YIELD

- ACEi/ARB first-line in proteinuric CKD: efferent dilation lowers intraglomerular pressure and proteinuria. S8; SG U5-R8
- Creatinine rise up to ~30% is expected; check K^+ and creatinine in 1–2 weeks; sick-day rules; never in pregnancy. S5; S8; S9; SG U5-R9

2 PART 2 · DRUG CLASS

Other Antihypertensives & Resistant Hypertension

SO3 IO5.1 IO5.2 IO5.3 IO5.4

CLASS

- **RAAS blockers are first-line — but rarely the whole story.** Other major classes: CCBs, beta-blockers, diuretics, central agents and direct vasodilators. S11; S11 (notes); S12 (notes)
- This four-card slide is a brief classification recap — CCBs and beta-blockers get fuller treatment later this session (angina) and in cardiovascular coursework. S12 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Calcium channel blockers: dihydropyridines (amlodipine) and non-dihydropyridines (diltiazem, verapamil). Beta-blockers: metoprolol, atenolol. Central α2 agonists: clonidine, methyldopa. Direct arteriolar vasodilator: hydralazine. S12; S28

MECHANISM OF ACTION

- Dihydropyridines (amlodipine): arteriolar dilation. Non-dihydropyridines (diltiazem, verapamil): also ↓ HR/contractility. S12
- Beta-blockers: ↓ heart rate, ↓ contractility, ↓ renin release. S12
- Chain (Figure 11): dihydropyridine arteriolar dilation → afterload falls; blood pressure lowers. Non-DHP CCB or beta-blocker: heart rate and contractility fall → renin release falls. S13; S13 (notes)
- Clonidine or methyldopa stimulates central α2 receptors → sympathetic outflow falls, so BP falls. Hydralazine dilates arterioles directly → reflex tachycardia and fluid retention. S16; S16 (notes)

INDICATIONS

- Beta-blockers: not generally first-line for uncomplicated HTN, but compelling indications include concurrent CAD/heart failure/arrhythmia. S12
- If all these classes lower blood pressure similarly, why are RAAS blockers preferred in CKD? Similar BP lowering; ACE inhibitors and ARBs add renoprotection — the renoprotective gap in Figure 12, not a difference in BP control. S14; S14 (notes)
- Thiazide / loop diuretics (covered in depth in Session 1): thiazides first-line with preserved renal function; loops preferred as GFR falls. S12
- Central agents (clonidine, methyldopa) and hydralazine: reserved for resistant hypertension or specific situations (e.g., methyldopa in pregnancy). S12; S16

- TB Diuretics enhance the efficacy of many agents, particularly ACE inhibitors; hydralazine and minoxidil need a diuretic because they cause salt and water retention. p. 280
- TB Spironolactone may be the most effective single agent for drug-resistant hypertension. p. 280

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- Peripheral edema is a CCB class adverse effect. S12
- Hydralazine: reflex tachycardia and fluid retention — must be paired with a beta-blocker and a diuretic. S16; S16 (notes)
- Abrupt clonidine withdrawal causes rebound hypertension. S16; S16 (notes)

CONTRAINDICATIONS

- Atenolol accumulates in renal impairment (reduce dose when CrCl is below 35 mL/min); beta-blockers: do not stop abruptly. S47; S48

DRUG INTERACTIONS

- Diltiazem or verapamil with a beta-blocker risks bradycardia and AV block; dihydropyridines pair with beta-blockers without that conduction risk. S13; S13 (notes)
- Verapamil and diltiazem raise digoxin levels. Rifampin and carbamazepine lower CCB levels; grapefruit juice raises dihydropyridine levels. S55; S55 (notes)

PHARMACOKINETICS

Not covered in materials

MONITORING

- **Resistant hypertension (BP above goal despite three or more antihypertensive classes, including a diuretic) is common in CKD, often driven by volume overload.** S15
- First step: confirm diuretic therapy is adequate — a thiazide may simply no longer be effective at the patient's current GFR (Session 1's loop-vs-thiazide cutoff). S15; S15 (notes)

- Switching from a thiazide to a loop diuretic, or increasing loop diuretic dosing, is often the most effective single change in advanced CKD. S15
- **Aldosterone antagonists (spironolactone) can be added for resistant hypertension, but require close potassium and renal-function monitoring, especially on top of an ACE inhibitor or ARB.** S15

CLINICAL PEARL / WHY

- "Resistant hypertension in CKD" is often really a diuretic-selection problem, not a need for an entirely new drug class — check diuretic adequacy first; thiazide to loop switch; add spironolactone with close monitoring. S15 (notes)
- Case: CKD (eGFR 28 mL/min), still hypertensive despite lisinopril, amlodipine, and hydrochlorothiazide. **What is the most likely contributor to inadequate control, and what change would you suggest first?** S17
- **At this GFR, hydrochlorothiazide has largely lost its natriuretic efficacy (Session 1). Switching to a loop diuretic such as furosemide or torsemide is usually the most effective next step before adding an entirely new drug class.** S17
- This case is a direct callback to the Session 1 "Choosing a Diuretic for Advancing CKD" case — point out the parallel if you remember it. S17 (notes)

Step by step

Resistant hypertension in CKD S15; S17

BP above goal on 3+ classes including a diuretic →

Check the diuretic first: thiazide at low GFR? →

Switch thiazide → loop, or raise the loop dose →

Then add spironolactone with close K^+ and renal monitoring

Figure 11 · CCBs and beta-blockers: different targets. Schematic (illustrative) — Figure 11. Vascular versus cardiac targets of antihypertensive drugs: dihydropyridines dilate arterioles (afterload falls); non-dihydropyridines and beta-blockers slow heart rate and contractility (beta-blockers also cut renin). S13; S13 (notes)

Figure 12 · BP-lowering efficacy vs. renoprotection in CKD. Original illustrative comparison (illustrative relative values, not from Katzung): BP-lowering efficacy is broadly comparable — ACEi/ARBs 70, CCBs 72, thiazide/loop diuretics 68, beta-blockers 55 — but renoprotection in proteinuric CKD is far greater with ACEi/ARBs (90) than CCBs (45), diuretics (35) or beta-blockers (30). S14; S14 (notes)

Figure 13 · Central agents and hydralazine. Schematic (illustrative) — Figure 13. Central agents and direct arteriolar vasodilation: clonidine and methyldopa reduce sympathetic outflow; hydralazine dilates arterioles directly, triggering reflex tachycardia and fluid retention. S16; S16 (notes)

Other major antihypertensive classes (S12)

Class	Examples	Mechanism	Key point
Calcium channel blockers	Amlodipine; diltiazem, verapamil	Arteriolar dilation; non-DHP also ↓ HR/contractility	Peripheral edema
Beta-blockers	Metoprolol, atenolol	↓ HR, contractility, renin	Compelling: CAD, HF, arrhythmia
Thiazide / loop diuretics	HCTZ; furosemide	Session 1	Loops as GFR falls
Central agents & direct vasodilators	Clonidine, methyldopa; hydralazine	Central α2; direct arteriolar	Resistant HTN; methyldopa in pregnancy

How to tell them apart. Sources: S12

Calcium channel blocker profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Amlodipine (Norvasc)	**Dihydropyridine calcium channel blocker**	Tablets 2.5/5/10 mg; oral suspension	Hypertension or angina: 5 mg PO daily, up to 10 mg daily; start 2.5 mg in older or frail patients	Hypertension, chronic stable angina, vasospastic angina
Nifedipine ER (Procardia XL, Adalat CC)	**Dihydropyridine calcium channel blocker**	Extended-release tablets 30/60/90 mg	30-60 mg PO daily, titrate to 90-120 mg daily; avoid immediate-release capsules for hypertension	Hypertension, chronic stable angina, vasospastic angina
Diltiazem (Cardizem)	**Non-dihydropyridine calcium channel blocker**	Tablets 30/60/90/120 mg; ER capsules 120-420 mg; IV injection	Hypertension: ER 120-240 mg PO daily up to 360 mg; angina: 30 mg QID (immediate-release) or ER daily	Hypertension, angina, atrial fibrillation rate control
Verapamil (Calan, Isoptin)	**Non-dihydropyridine calcium channel blocker**	Tablets 40/80/120 mg; ER tablets 120/180/240 mg; IV injection	Hypertension: SR 180-240 mg PO daily up to 480 mg; angina: 80-120 mg TID	Hypertension, angina, atrial fibrillation and SVT rate control

How to tell them apart. Sources: S43; S44; S45; S46

Calcium channel blocker profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Amlodipine (Norvasc)	Peripheral edema, flushing, headache, dizziness, gingival hyperplasia	Hypersensitivity, severe aortic stenosis (caution), cardiogenic shock	Profound hypotension with reflex tachycardia, prolonged vasodilation; slow onset and long half-life
Nifedipine ER (Procardia XL, Adalat CC)	Peripheral edema, headache, flushing, reflex tachycardia, constipation	Cardiogenic shock, hypersensitivity; avoid immediate-release forms in acute coronary syndrome	Severe hypotension, reflex tachycardia, edema; swallow ER tablet whole
Diltiazem (Cardizem)	Bradycardia, AV block, edema, constipation, dizziness, CYP3A4 interactions	Sick sinus syndrome, second- or third-degree AV block, severe hypotension, acute MI with pulmonary congestion, reduced EF	Bradycardia, AV block, heart failure exacerbation; strongly interacts with beta-blockers
Verapamil (Calan, Isoptin)	Constipation, bradycardia, AV block, edema, dizziness, gingival hyperplasia	Sick sinus syndrome, AV block, severe LV dysfunction, WPW with atrial fibrillation, cardiogenic shock	Severe bradycardia, AV block, heart failure; avoid with beta-blockers; raises digoxin levels

How to tell them apart. Sources: S43; S44; S45; S46

Beta-blocker profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Metoprolol (Lopressor, Toprol-XL)	**Beta-1 selective beta-blocker**	Tartrate tablets 25/50/100 mg; succinate ER tablets 25-200 mg; IV injection	Hypertension: tartrate 50 mg PO BID or succinate 50-100 mg daily (maximum 400 mg/day); heart failure: succinate 12.5-25 mg daily, titrate	Hypertension with CAD or HF, angina, post-MI, atrial fibrillation
Atenolol (Tenormin)	**Beta-1 selective beta-blocker**	Tablets 25/50/100 mg	Hypertension: 25-50 mg PO daily (maximum 100 mg); reduce dose when CrCl is below 35 mL/min	Hypertension, angina, post-MI

How to tell them apart. Sources: S47; S48

Beta-blocker profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Metoprolol (Lopressor, Toprol-XL)	Fatigue, bradycardia, dizziness, cold extremities, sexual dysfunction, bronchospasm at high dose	Severe bradycardia, high-grade AV block, decompensated HF, cardiogenic shock; do not stop abruptly	Bradycardia, hypotension, heart block, bronchospasm; rebound tachycardia with abrupt withdrawal
Atenolol (Tenormin)	Fatigue, bradycardia, cold extremities, depression, erectile dysfunction	Severe bradycardia, high-grade AV block, decompensated HF; do not stop abruptly	Accumulates in renal impairment, causing bradycardia and hypotension; rebound after abrupt withdrawal

How to tell them apart. Sources: S47; S48

Central agent and hydralazine profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Clonidine (Catapres)	**Central alpha-2 agonist**	Tablets 0.1/0.2/0.3 mg; transdermal patch 0.1-0.3 mg/day weekly	Hypertension: 0.1 mg PO BID, titrate (usual maximum 2.4 mg/day); patch 0.1 mg/day weekly	Resistant hypertension, hypertensive urgency adjunct, opioid withdrawal adjunct
Methyldopa (Aldomet)	**Central alpha-2 agonist**	Tablets 250/500 mg	250 mg PO BID-TID, titrate (maximum 3 g/day); preferred oral option for hypertension in pregnancy	Hypertension in pregnancy, resistant hypertension

Drug	Class	Formulation	Adult dose	Indications
Hydralazine (Apresoline)	**Direct arteriolar vasodilator**	Tablets 10/25/50/100 mg; IV injection	Hypertension: 10 mg PO QID, titrate to 25-50 mg QID (maximum 300 mg/day); heart failure with isosorbide dinitrate: 25-75 mg TID	Resistant hypertension, hypertensive emergency in pregnancy, heart failure (with nitrate)

How to tell them apart. Sources: S49; S50; S51

Central agent and hydralazine profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Clonidine (Catapres)	Sedation, dry mouth, bradycardia, constipation, rebound hypertension	Hypersensitivity; use caution in bradycardia and AV block	Rebound hypertension with abrupt withdrawal; overdose causes bradycardia, hypotension, coma and miosis
Methyldopa (Aldomet)	Sedation, depression, dry mouth, positive Coombs test, hepatotoxicity	Active liver disease, concomitant MAO inhibitors, history of methyldopa hepatitis	Hemolytic anemia, hepatitis, drug fever; monitor liver tests and CBC
Hydralazine (Apresoline)	Headache, reflex tachycardia, fluid retention, flushing, drug-induced lupus	Coronary artery disease without beta-blocker, mitral valve rheumatic disease, hypersensitivity	Reflex tachycardia, angina, lupus-like syndrome with high dose or slow acetylators; pair with beta-blocker and diuretic

How to tell them apart. Sources: S49; S50; S51

WHY IT HAPPENS

- Why pair hydralazine with a beta-blocker and a diuretic? Direct dilation triggers reflex tachycardia and fluid retention; the partners cancel both. S16

HIGH-YIELD

- Non-DHP CCB + beta-blocker → bradycardia/AV block; DHP + beta-blocker is fine. S13
- Resistant HTN in CKD: fix the diuretic first (thiazide → loop), then spironolactone with K^+ monitoring. S15; S17

3 PART 3 · DRUG CLASS

Nitrates, Angina & the Triple Whammy

SO4 SO5 IO5.1 IO5.2 IO5.3 IO5.6

CLASS

- **Where cardiovascular and renal pharmacology converge — and where that convergence can go wrong.** This section covers vasodilators and angina, then the cardiorenal overlap. S18; S18 (notes)
- **Figure 15. Original illustrative diagram (not from Katzung): angina results when myocardial oxygen demand exceeds supply; nitrates reduce preload, dihydropyridine CCBs reduce afterload,** S21
- **non-dihydropyridine CCBs also reduce heart rate/contractility and increase coronary flow, and beta-blockers reduce heart rate and contractility.** S21
- This figure organizes every antianginal drug class by exactly which side of the supply/demand equation it affects — sort nitrates, each CCB subtype, and beta-blockers onto it from memory. S21 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Nitroglycerin (sublingual for acute angina; IV/patch for sustained use), isosorbide mononitrate/dinitrate (oral, longer-acting). S19

MECHANISM OF ACTION

- **Mechanism: release nitric oxide, activating guanylate cyclase and increasing cGMP, producing smooth muscle relaxation — predominantly venodilation at usual doses, reducing preload and myocardial oxygen demand.** S19; SG U5-R11
- Chain (Figure 14): nitrate donates nitric oxide (NO) → NO activates guanylate cyclase, so cGMP rises → cGMP relaxes smooth muscle: venodilation → preload and oxygen demand fall. S20; S20 (notes)

INDICATIONS

- CCBs are the clearest example in the unit of the same drug class, same mechanism, treating two different conditions (hypertension and angina). S22; SG U5-R10
- This is the deliberate cardiovascular-renal "overlap" moment the lecture title promises — calcium channel blockers are not a separate topic from the antihypertensive content in Section 5; they are the same drugs doing double duty. S22 (notes)

- **Dihydropyridines (amlodipine, nifedipine) — predominantly arteriolar dilation, reducing afterload; the same mechanism used for hypertension (Section 5).** S22; SG U5-R10
- **Non-dihydropyridines (diltiazem, verapamil) — also reduce heart rate and contractility and increase coronary blood flow; avoid combining with beta-blockers due to additive bradycardia/AV block risk.** S22

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- **Adverse effects: headache (vasodilation), reflex tachycardia, orthostatic hypotension.** S19
- **Adverse effects: peripheral edema and reflex tachycardia (dihydropyridines); bradycardia, AV block, and constipation (non-dihydropyridines, especially verapamil).** S22

CONTRAINDICATIONS

- **Absolute contraindication: concurrent use with PDE5 inhibitors (sildenafil and similar) — the combination can cause life-threatening hypotension.** S19; SG U5-R11
- PDE5 inhibitors such as sildenafil block cGMP breakdown, causing severe hypotension with nitrates. S20; S20 (notes)
- The PDE5-inhibitor interaction is one of the highest-stakes medication-safety points in this whole unit — a real-world, repeatedly tested safety question worth a memorable callout. S19 (notes)

DRUG INTERACTIONS

- **Figure 16. Original illustrative diagram (not from Katzung): an NSAID (afferent arteriolar constriction), an ACE inhibitor or ARB (efferent arteriolar dilation), and a diuretic (volume depletion) each independently reduce glomerular filtration pressure —** S23; SG U5-R12
- **combined, the effect on kidney perfusion is additive to synergistic, substantially raising acute kidney injury risk.** An extremely common, often missed, entirely preventable cause of AKI. S23; S23 (notes); SG U5-R12

PHARMACOKINETICS

Not covered in materials

- Three hits (Figure 17): NSAID constricts the afferent arteriole → ACE inhibitor or ARB dilates the efferent arteriole → diuretic causes volume depletion → filtration pressure collapses: AKI. S24; S24 (notes)
- TB NSAIDs blunt loop and thiazide action by reducing renal prostaglandins — significant in nephrotic syndrome or cirrhosis. p. 270; p. 272

MONITORING

- Tolerance develops with continuous exposure — a nitrate-free interval (typically overnight) is required to maintain efficacy with long-acting formulations. S19; S20
- Prevention: counsel patients about over-the-counter NSAIDs — ask which OTC NSAIDs they commonly take. S24; S24 (notes)

CLINICAL PEARL / WHY

- The triple whammy is the capstone cardiorenal interaction: Session 1's diuretics + this session's RAAS blockers + NSAIDs students will meet constantly in practice. S23 (notes)
- Hypertension → CKD: uncontrolled blood pressure causes glomerular capillary injury and nephrosclerosis over time — a leading cause of CKD progression, alongside diabetes. S25; S25 (notes)
- CKD → hypertension: reduced renal function causes volume overload and RAAS activation, which worsens and complicates blood pressure control, often requiring multiple drug classes. S25; S25 (notes)
- **Figure 18. Original illustrative diagram (not from Katzung): hypertension and chronic kidney disease drive one another in a self-reinforcing cycle — uncontrolled blood pressure causes glomerular injury and CKD progression,** S26
- **while CKD causes volume overload and RAAS activation that worsens hypertension.** S26
- Figure 18 is the conceptual closer for the whole unit — tie it back to Figure 8: RAAS blockade is effective precisely because it intervenes in both directions of this cycle at once. S26 (notes)
- Case: an older adult on lisinopril and hydrochlorothiazide starts a regular OTC NSAID for a knee injury; one week later, lethargy and an acute rise in creatinine. **What is the most likely explanation, and how could this have been prevented?** S27
- **This is a classic "triple whammy" presentation (Figure 16) — the NSAID, ACE inhibitor, and diuretic each independently reduced glomerular filtration pressure, and together precipitated acute kidney injury.** S27; SG U5-S
- **Prevention requires counseling patients on RAAS blockers and diuretics to avoid regular NSAID use and to seek guidance before starting one, even an over-the-counter product.** S27; SG U5-S

- This case is the direct clinical application of Figure 16 and reinforces IO5.6 (patient counseling): the preventable element is entirely what the patient was or wasn't told about OTC NSAID use. S27 (notes)
- Same patient, eGFR now 25: hydrochlorothiazide has likely lost much of its efficacy — switching to a loop diuretic (furosemide, torsemide) is more effective for blood pressure regardless of the NSAID issue. SG U5-S
- Counsel the patient to avoid regular OTC NSAID use and to check with a prescriber or pharmacist before starting any new OTC product, given the RAAS blocker + diuretic combination already in place. SG U5-S

Step by step

Triple whammy S24; S27

NSAID constricts the afferent arteriole →

ACE inhibitor or ARB dilates the efferent arteriole →

Diuretic causes volume depletion → Filtration pressure collapses → AKI →

Prevent: counsel on OTC NSAIDs

Figure 14 · Nitrates: NO, cGMP and venodilation. Schematic (illustrative) — Figure 14. Nitrate action: NO → guanylate cyclase → cGMP → smooth-muscle relaxation (venodilation) → lower preload and oxygen demand. PDE5 inhibitors block cGMP breakdown — severe hypotension with nitrates. S20; S20 (notes)

Figure 15 · Angina: where nitrates and CCBs act. Original illustrative diagram (not from Katzung): angina results when myocardial oxygen demand exceeds supply. Nitrates reduce preload; dihydropyridine CCBs reduce afterload; non-dihydropyridine CCBs also reduce heart rate/contractility and increase coronary flow; beta-blockers reduce heart rate and contractility. S21; S21 (notes)

Figure 16 · The triple whammy. Original illustrative diagram (not from Katzung): NSAID (afferent constriction) + ACE inhibitor or ARB (efferent dilation) + diuretic (volume depletion) each independently lower glomerular filtration pressure — combined, additive to synergistic AKI risk. S23; S23 (notes)

Figure 17 · Three hits on glomerular pressure. Schematic (illustrative) — Figure 17. Three hits on glomerular filtration pressure: afferent constriction (NSAID), efferent dilation (ACE inhibitor or ARB), volume depletion (diuretic) → filtration pressure collapses: AKI. Prevention: counsel patients about OTC NSAIDs. S24; S24 (notes)

Figure 18 · The cardiorenal-hypertension cycle. Original illustrative diagram (not from Katzung): uncontrolled blood pressure causes glomerular injury and CKD progression, while CKD causes volume overload and RAAS activation that worsens hypertension — a self-reinforcing cycle that RAAS blockade interrupts in both directions. S26; S26 (notes)

Calcium channel blockers: DHP vs. non-DHP (S22)

	Dihydropyridines	Non-dihydropyridines
Examples	Amlodipine, nifedipine	Diltiazem, verapamil
Main action	Arteriolar dilation → ↓ afterload	↓ HR, contractility; ↑ coronary flow
Adverse effects	Peripheral edema, reflex tachycardia	Bradycardia, AV block, constipation (verapamil)
With a beta-blocker	Safe pairing	Avoid — bradycardia/AV block

How to tell them apart. Sources: S13; S22

Drug names reveal the class: cardiorenal agents (S28)

Stem	Class	Examples	Memory aid
-pril	ACE inhibitor	Lisinopril, enalapril, captopril	Angiotensin-converting enzyme
-sartan	ARB	Losartan, valsartan	Angiotensin receptor blocker
-kiren	Direct renin inhibitor	Aliskiren	Renin is the top of the cascade
-dipine	Dihydropyridine CCB	Amlodipine, nifedipine	Arteriolar dilation

Stem	Class	Examples	Memory aid
-olol	Beta-blocker	Metoprolol, atenolol	Heart rate and renin down
-azine / -alazine	Hydralazine vasodilator	Hydralazine	Pair with beta-blocker and diuretic
-onidine	Central alpha-2 agonist	Clonidine	Central, so sympathetic outflow falls
-nitrate / nitro-	Nitrates	Isosorbide, nitroglycerin	Venodilation, preload down
-afil	PDE5 inhibitor	Sildenafil	Never combine with nitrates
-semide	Loop diuretic	Furosemide	Loop of Henle

How to tell them apart. Sources: S28; S28 (notes)

Supporting drugs at a glance (S55)

Drug(s)	Why it matters here
NSAIDs	Blunt antihypertensives; raise AKI risk with ACE inhibitor or ARB plus diuretic
PDE5 inhibitors (such as sildenafil)	Never combine with nitrates
Potassium supplements and potassium-sparing diuretics	Hyperkalemia with ACE inhibitors and ARBs
Trimethoprim	Hyperkalemia
Loop and thiazide diuretics	Covered in Session 1
Statins and aspirin	Cardiovascular risk reduction
Digoxin	Verapamil and diltiazem raise digoxin levels
Rifampin and carbamazepine	Lower calcium channel blocker levels
Grapefruit juice	Raises dihydropyridine calcium channel blocker levels

How to tell them apart. Sources: S55; S55 (notes)

Nitrate profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Nitroglycerin (Nitrostat, Nitro-Dur)	**Organic nitrate**	Sublingual tablet 0.3/0.4/0.6 mg; spray 0.4 mg; patch 0.1-0.8 mg/h; IV infusion	Acute angina: 0.4 mg SL every 5 minutes up to 3 doses; patch 0.2-0.8 mg/h applied 12-14 hours per day; IV 5-10 mcg/min titrated	Acute angina, angina prophylaxis, acute heart failure, hypertensive emergency

Drug	Class	Formulation	Adult dose	Indications
Isosorbide mononitrate (Imdur)	**Organic nitrate (long-acting)**	ER tablets 30/60/120 mg; immediate-release tablets 10/20 mg	ER: 30-60 mg PO every morning, up to 240 mg/day; immediate-release: 20 mg BID given 7 hours apart	Angina prophylaxis
Isosorbide dinitrate (Isordil)	**Organic nitrate**	Tablets 5/10/20/30/40 mg; ER capsules 40 mg; combination with hydralazine (BiDil)	Angina: 10-40 mg PO TID with a 14-hour nitrate-free interval; heart failure: 20-40 mg TID with hydralazine	Angina prophylaxis, heart failure in Black patients (with hydralazine)

How to tell them apart. Sources: S52; S53; S54

Nitrate profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Nitroglycerin (Nitrostat, Nitro-Dur)	Headache, flushing, hypotension, reflex tachycardia, dizziness	Concurrent PDE5 inhibitors (sildenafil, tadalafil) or riociguat, severe anemia, increased intracranial pressure, right ventricular infarction	Severe hypotension and syncope, especially with PDE5 inhibitors; tolerance without a nitrate-free interval; methemoglobinemia at very high doses
Isosorbide mononitrate (Imdur)	Headache, dizziness, hypotension, flushing	Concurrent PDE5 inhibitors or riociguat, severe anemia, increased intracranial pressure	Hypotension, headache; tolerance requires nitrate-free interval
Isosorbide dinitrate (Isordil)	Headache, flushing, hypotension, reflex tachycardia	Concurrent PDE5 inhibitors or riociguat, severe anemia, increased intracranial pressure	Hypotension, syncope, headache; tolerance without a nitrate-free interval

How to tell them apart. Sources: S52; S53; S54

WHY IT HAPPENS

- Why are nitrates + sildenafil deadly? Nitrates make cGMP; sildenafil stops its breakdown — cGMP piles up and blood pressure collapses. S20

MNEMONIC (MEMORY AID)

- Stems: **-pril**, **-sartan**, **-kiren**, **-dipine**, **-olol**, **-azine / -alazine**, **-onidine**, **-nitrate / nitro-**, **-afil**, **-semide**. Read the **Stem**, name the **Class**, recall the **Examples** and the **Memory aid**. S28; S28 (notes)

HIGH-YIELD

- Nitrates: NO → cGMP → venodilation, ↓ preload; nitrate-free interval; never with PDE5 inhibitors. S19; S20; SG U5-R11

- Triple whammy: NSAID (afferent) + ACEi/ARB (efferent) + diuretic (volume) → AKI; counsel on OTC NSAIDs. S23; S27; SG U5-R12

4 PART 4 · TOPIC

Putting It All Together: Renal Toolkit

SO5 IO5.3 IO5.4 IO5.6

KEY DEFINITIONS

- **Applying the whole Renal pharmacology toolkit.** Integration: put the drug classes together — for each patient, name a first-line approach AND the single biggest safety concern to check before prescribing or counseling. S29; S29 (notes); S30 (notes)

HOW IT WORKS

- **1.** Diuretic classes (Session 1) and RAAS-blocking antihypertensives both map onto specific, drug-targetable points in renal physiology — the nephron (Figure 1) and the RAAS cascade (Figure 8), respectively. S31
- **2.** ACE inhibitors and ARBs are preferred first-line therapy for hypertension in CKD because of their renoprotective efferent-arteriolar mechanism, not just their blood-pressure-lowering effect. S31
- **3.** Calcium channel blockers exemplify the cardiovascular-renal overlap directly — the same drug class, same mechanism, treats both hypertension and angina. S31
- **4.** The "triple whammy" (NSAID + ACEi/ARB + diuretic) is a classic, common, and preventable cause of acute kidney injury that draws on content from both sessions at once. S31
- **5.** Hypertension and chronic kidney disease exist in a bidirectional, self-reinforcing cycle (Figure 18) — which is exactly why managing one well so often means managing the other. S31
- All three threads center on glomerular filtration pressure. S31 (notes)
- Connecting back to earlier units: drug-metabolism/dosing concepts (Unit 1) and the biologic-mechanism callbacks from Units 2–4 all reinforce that pharmacology is cumulative — the cardiorenal interface is where several organ-system units visibly intersect. S31 (notes)

DRUGS NAMED AS EXAMPLES

- **A cumulative self-test spanning both sessions — use this to review before exams.** The recall quiz is a standalone resource, separate from in-lecture quick checks. S32; S32 (notes); S35 (notes)
- **Recall Q1–2:** loops collapse the lumen-positive potential (↑ Ca^{2+} excretion), thiazides increase distal Ca^{2+} reabsorption; at eGFR 22 switch the thiazide to a loop. S33; S35
- **Recall Q3–4:** a creatinine rise up to ~30% on an ACE inhibitor is expected — continue with monitoring; ACEi/ARBs dilate the efferent arteriole, lowering intraglomerular pressure and proteinuria. S33; S34; S35
- **Recall Q5:** the triple whammy is three independent reductions in filtration pressure combining additively to synergistically, precipitating AKI. S34; S35
- **Recall Q6:** CCBs treat hypertension via arteriolar dilation (reducing afterload/systemic vascular resistance) and angina via the same mechanism (reducing afterload and myocardial oxygen demand) — one drug class, one mechanism, two indications. S34; S35

CLINICAL

- **Patient A — advancing CKD, hypertension no longer controlled on a thiazide:** switch to a loop diuretic; check potassium and volume status. S30; S15; S17
- **Patient B — proteinuric CKD, needs first-line antihypertensive therapy:** ACE inhibitor or ARB; check potassium and creatinine in 1–2 weeks and pregnancy potential. S30; S5; S8
- **Patient C — on an ACE inhibitor and a diuretic, about to start a daily OTC NSAID:** counsel against regular NSAID use — triple whammy AKI risk. S30; S27
- **Patient D — stable angina, already on a beta-blocker, needs additional symptom control:** add a dihydropyridine CCB or a long-acting nitrate; avoid diltiazem/verapamil (AV block) and PDE5 inhibitors with nitrates. S30; S13; S19; S22

TRAP

- Reflexively stopping an ACE inhibitor for a ≤30% creatinine rise is the classic trap. S5 (notes); S10 (notes)

- Adding a new drug class when the real problem is a thiazide at low GFR. S15 (notes); S17

Which approach would you choose? (S30)

Patient	First-line approach	Biggest safety check
A · Advancing CKD, BP no longer controlled on a thiazide	Switch to a loop diuretic	Potassium, volume status
B · Proteinuric CKD, first-line therapy	ACE inhibitor or ARB	K^+ and creatinine in 1–2 weeks; pregnancy
C · ACEi + diuretic, about to start daily OTC NSAID	Counsel to avoid the NSAID	Triple whammy AKI
D · Stable angina on a beta-blocker	Add DHP CCB or long-acting nitrate	No non-DHP with BB; no PDE5 with nitrates

How to tell them apart. Sources: S30; S13; S15; S19; S27

MNEMONIC (MEMORY AID)

- Every case, two answers: **the approach** and **the one thing to check**. S30 (notes)

HIGH-YIELD

- Five takeaways: two targetable maps · ACEi/ARB renoprotection · CCB overlap · triple whammy · HTN–CKD cycle. S31

Cram sheet

Every term, mechanism, drug, and number. Hide the answers and quiz yourself; tap a blurred answer to peek.

Prompt	Answer
RAAS blocks	Aliskiren renin · -pril ACE · -sartan AT1 · spironolactone aldosterone S4; S28; SG U5-R7
ACEi adverse effects	Cough (bradykinin), angioedema, hyperkalemia S5; S6
ACEi cough → ?	Switch to an ARB S6; S7
ACEi/ARB in pregnancy	Contraindicated (fetal renal toxicity) S5; S7
Why first-line in proteinuric CKD	Efferent dilation → ↓ intraglomerular pressure, ↓ proteinuria S8; SG U5-R8
Creatinine after ACEi	Up to ~30% expected; more → stenosis or volume depletion S5; S10; SG U5-R9
Monitoring	K^+ and creatinine within 1–2 weeks; sick-day rules S8; S9
Aliskiren	Never with ACEi/ARB in diabetes S7; S42
DHP vs. non-DHP	Arterioles · heart rate/contractility S12; S22
Non-DHP + beta-blocker	Bradycardia, AV block S13
Hydralazine	Pair with beta-blocker + diuretic S16
Clonidine	Rebound hypertension if stopped abruptly S16
Methyldopa	Hypertension in pregnancy S16; S50
Nitrates	NO → cGMP → venodilation; nitrate-free interval S19; S20
Nitrate + PDE5 inhibitor	Absolute contraindication — hypotension S19; SG U5-R11
HTN and angina, one mechanism	Calcium channel blockers S22; SG U5-R10
Triple whammy	NSAID + ACEi/ARB + diuretic S23; SG U5-R12
Once-daily ACE inhibitor	Levels stay far above C50 p. 50
Hydralazine + diuretic	Vasodilators cause salt and water retention p. 280

Self-test

First, the lecture's own 6 knowledge checks — exactly as written. Then original PANCE-style questions. Tap an answer to reveal the explanation.

From the lecture

Lecture check 1 · S33; S35

Why do loop diuretics increase calcium excretion while thiazide diuretics decrease it?

- **A** Loop diuretics act in the collecting duct; thiazides act in the proximal tubule
- **B** Loop diuretics collapse the lumen-positive potential that drives paracellular calcium reabsorption; thiazides do not
- **C** Thiazides block the Na-K-2Cl cotransporter
- **D** There is no mechanistic difference between the two classes

ANSWER: B

Loop diuretics block NKCC2 in the thick ascending limb, collapsing the lumen-positive potential that normally drives paracellular Ca^{2+}/Mg^{2+} reabsorption, so calcium excretion rises; thiazides act at the distal tubule and, through a separate mechanism, increase distal calcium reabsorption instead. S33; S35

Lecture check 2 · S33; S35

A patient's thiazide diuretic stops controlling blood pressure as CKD advances to an eGFR of 22 mL/min. What is the most appropriate change?

- **A** Increase the thiazide dose
- **B** Switch to a loop diuretic
- **C** Add a second thiazide
- **D** Discontinue diuretic therapy entirely

ANSWER: B

Thiazide diuretics lose most of their natriuretic efficacy once GFR falls below roughly 30 mL/min. Loop diuretics act upstream at the thick ascending limb and remain effective in advanced CKD. S33; S35

Lecture check 3 · S33; S35

A patient starts lisinopril for proteinuric CKD and the serum creatinine rises by 25% within two weeks. What is the most appropriate response?

- **A** Stop the ACE inhibitor immediately
- **B** Continue therapy with ongoing monitoring, since this rise is expected
- **C** Double the dose
- **D** Switch to an NSAID instead

ANSWER: B

A creatinine rise of up to approximately 30% after starting an ACE inhibitor or ARB reflects reduced intraglomerular pressure from efferent arteriolar dilation — an expected hemodynamic effect, not toxicity — so therapy should generally continue with monitoring. S33; S35

Lecture check 4 · S34; S35

Why are ACE inhibitors and ARBs preferred first-line therapy for hypertension in proteinuric CKD?

- **A** They preferentially dilate the efferent arteriole, lowering intraglomerular pressure and proteinuria
- **B** They have no renal effects beyond lowering systemic blood pressure
- **C** They are the only antihypertensive class that can be used in CKD
- **D** They increase aldosterone release to protect sodium balance

ANSWER: A

ACE inhibitors and ARBs preferentially dilate the efferent arteriole, lowering intraglomerular pressure and reducing proteinuria — a structural renoprotective effect beyond systemic blood pressure control, which is why they are preferred in proteinuric CKD specifically. S34; S35

Lecture check 5 · S34; S35

A patient on an ACE inhibitor and a diuretic starts a daily OTC NSAID and develops acute kidney injury. What is the mechanism?

- **A** Three independent reductions in glomerular filtration pressure combining additively
- **B** NSAIDs have no renal effects
- **C** The ACE inhibitor directly damages the NSAID
- **D** This combination has no increased risk

ANSWER: A

This is the "triple whammy": the NSAID constricts the afferent arteriole, the ACE inhibitor dilates the efferent arteriole, and the diuretic causes volume depletion — three independent reductions in glomerular filtration pressure that combine additively to synergistically, precipitating AKI. S34; S35

Lecture check 6 · S34; S35

Which drug class is used for BOTH hypertension and angina pectoris via the same underlying mechanism?

- **A** Thiazide diuretics
- **B** Calcium channel blockers
- **C** Carbonic anhydrase inhibitors
- **D** Aldosterone antagonists

ANSWER: B

Calcium channel blockers treat hypertension via arteriolar dilation (reducing afterload/systemic vascular resistance) and treat angina via the same arteriolar-dilation mechanism (reducing afterload and myocardial oxygen demand) — the clearest example of one drug class, one mechanism, two indications in this unit. S34; S35

Original questions

Q1 · Mechanism · RAAS Blockers

Why does an ACE inhibitor cause a dry cough?

- **A** Aldosterone excess dries the airway
- **B** It blocks AT1 receptors in the lung
- **C** Angiotensin II rises and irritates airways
- **D** Bradykinin accumulates because ACE also degrades it
- **E** It is an IgE-mediated allergy to the drug

ANSWER: D

ACE also breaks down bradykinin; blocking it raises bradykinin, causing cough and angioedema (S6). S6; S7

- **A.** Aldosterone falls (S5).
- **B.** That is an ARB, which spares cough (S7).
- **C.** Angiotensin II falls (S5).
- **E.** Cough is not allergic (S6).

Q2 · Clinical vignette · RAAS Blockers

A woman of reproductive age with proteinuric CKD needs antihypertensive therapy. What must be addressed first with an ACE inhibitor?

- **A** Rebound hypertension if stopped
- **B** Bradykinin cough risk only
- **C** Pregnancy risk — fetal renal toxicity
- **D** Ototoxicity with loop diuretics
- **E** Gynecomastia with long-term use

ANSWER: C

ACE inhibitors (and ARBs) are contraindicated in pregnancy; avoid in women who are or may become pregnant (S5; S7). S5; S7

- **A.** Clonidine (S16).
- **B.** Pregnancy is the contraindication (S5).
- **D.** Not an ACEi issue (L09).
- **E.** Spironolactone (L09).

Q3 · Mechanism · RAAS Blockers

Why are ACE inhibitors and ARBs preferred in proteinuric CKD beyond lowering blood pressure?

- **A** They raise GFR by expanding volume
- **B** They increase aldosterone to retain sodium
- **C** They constrict the afferent arteriole
- **D** They dilate the efferent arteriole
- **E** They are the only class allowed in CKD

ANSWER: D

Reducing angiotensin II preferentially dilates the efferent arteriole, lowering intraglomerular pressure and proteinuria (S8; SG U5-R8). S8; S9; SG U5-R8

- **A.** Not the mechanism (S8).
- **B.** They lower aldosterone (S5).
- **C.** That is NSAIDs (S23).
- **E.** Other classes are used too (S12).

Q4 · Clinical vignette · Other Antihypertensives & Resistant Hypertension

A patient on metoprolol develops heart block after a second drug is added for blood pressure. Which drug most likely?

- **A** Losartan
- **B** Lisinopril
- **C** Verapamil
- **D** Amlodipine
- **E** Hydralazine

ANSWER: C

Non-dihydropyridines (diltiazem, verapamil) with a beta-blocker risk bradycardia and AV block (S13). S13; S46

- **A.** No conduction effect (S7).
- **B.** No conduction effect (S5).
- **D.** DHPs pair safely with beta-blockers (S13).
- **E.** Causes reflex tachycardia (S16).

Q5 · Clinical vignette · Other Antihypertensives & Resistant Hypertension

A pregnant patient needs an oral antihypertensive. Which central agent is the preferred option on the slides?

- **A** Aliskiren
- **B** Methyldopa
- **C** Losartan
- **D** Clonidine
- **E** Lisinopril

ANSWER: B

Methyldopa is used in pregnancy (S12; S16; S50). S12; S16; S50

- **A.** Contraindicated in pregnancy (S42).
- **C.** Contraindicated in pregnancy (S7).
- **D.** Rebound risk; not the pregnancy agent (S16).
- **E.** Contraindicated in pregnancy (S5).

Q6 · Mechanism · Other Antihypertensives & Resistant Hypertension

Why must hydralazine be paired with a beta-blocker and a diuretic?

- **A** To reduce its hepatic first-pass loss
- **B** To prevent bradykinin cough
- **C** To avoid hyperkalemia from RAAS blockade
- **D** To block reflex tachycardia and fluid retention
- **E** To prevent rebound hypertension on stopping

ANSWER: D

Direct arteriolar dilation triggers reflex tachycardia and fluid retention (S16). S16

- **A.** Not the taught reason (S16).
- **B.** That is ACE inhibitors (S6).
- **C.** Not the reason (S16).
- **E.** That is clonidine (S16).

Q7 · Mechanism · Other Antihypertensives & Resistant Hypertension

What happens if clonidine is stopped abruptly?

- **A** Angioedema
- **B** Drug-induced lupus
- **C** Bradykinin cough
- **D** Hyperkalemia
- **E** Rebound hypertension

ANSWER: E

Abrupt clonidine withdrawal causes rebound hypertension (S16; S49). S16; S49

- **A.** ACE inhibitors (S5).
- **B.** Hydralazine (S51).
- **C.** ACE inhibitors (S6).
- **D.** RAAS blockers (S5).

Q8 · Clinical vignette · Nitrates, Angina & the Triple Whammy

A man using sildenafil asks for a nitroglycerin prescription for chest pain. What is the key concern?

- **A** Severe hyperkalemia
- **B** Faster nitrate tolerance only
- **C** Life-threatening hypotension
- **D** Bradykinin-mediated cough
- **E** Reflex bradycardia and AV block

ANSWER: C

Concurrent PDE5 inhibitors are an absolute contraindication to nitrates — severe hypotension (S19–S20). S19; S20; SG U5-R11

- **A.** Not a nitrate effect (S19).
- **B.** The interaction is the danger (S19).
- **D.** ACE inhibitors (S6).
- **E.** Nitrates cause reflex tachycardia (S19).

Q9 · Mechanism · Nitrates, Angina & the Triple Whammy

How do nitrates relieve angina?

- **A** They inhibit PDE5 to raise cAMP
- **B** NO raises cGMP, venodilation cuts preload
- **C** They block β1 receptors to slow the heart
- **D** They block L-type calcium channels
- **E** They block ACE to lower angiotensin II

ANSWER: B

Nitrates release NO → guanylate cyclase → cGMP → venodilation, reducing preload and oxygen demand (S19–S20). S19; S20

- **A.** PDE5 inhibitors (S20).
- **C.** Beta-blockers (S21).
- **D.** CCBs (S22).
- **E.** ACE inhibitors (S5).

Q10 · Mechanism · Nitrates, Angina & the Triple Whammy

Why does a patient on long-acting isosorbide mononitrate need a nitrate-free interval?

- **A** Continuous exposure causes tolerance
- **B** To clear its toxic metabolite
- **C** To let potassium normalize
- **D** To prevent angioedema
- **E** To prevent rebound hypertension

ANSWER: A

Tolerance develops with continuous exposure; a nitrate-free interval (typically overnight) maintains efficacy (S19). S19; S53

- **B.** Not the reason (S19).
- **C.** Not a nitrate issue (S19).
- **D.** ACE inhibitors (S5).
- **E.** Clonidine (S16).

Q11 · Mechanism · Nitrates, Angina & the Triple Whammy

Which drug class is used for both hypertension and angina via the same mechanism?

- **A** Thiazide diuretics
- **B** Carbonic anhydrase inhibitors
- **C** Calcium channel blockers
- **D** Aldosterone antagonists
- **E** Central alpha-2 agonists

ANSWER: C

CCBs dilate arterioles, lowering afterload — treating both hypertension and angina (S22; SG U5-R10). S22; SG U5-R10

- **A.** Hypertension only (L09).
- **B.** Niche diuretic uses (L09).
- **D.** Not an angina class (L09).
- **E.** Resistant HTN (S16).

Objectives checklist

Tick only what you could teach out loud. Saved on this device.

- [ ] **IO5.1** Identify major pharmacologic classes used in the management of renal conditions and describe their mechanisms of action.
- [ ] **IO5.2** Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat renal conditions.
- [ ] **IO5.3** Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.
- [ ] **IO5.4** Select appropriate pharmacologic therapy for common renal conditions based on clinical presentation and patient-specific factors.
- [ ] **IO5.5** Identify appropriate dosing, administration, and monitoring considerations for medications used in renal conditions.
- [ ] **IO5.6** Explain important medication-use and safety information to patients receiving pharmacotherapy for renal.
- [ ] **SO1** Describe the renin-angiotensin-aldosterone system (RAAS) and identify where each antihypertensive drug class acts.
- [ ] **SO2** Explain why ACE inhibitors and ARBs are preferred first-line therapy for hypertension in chronic kidney disease, including their renoprotective mechanism.
- [ ] **SO3** Identify major antihypertensive drug classes beyond RAAS blockers, their mechanisms, and their role in CKD-specific management.
- [ ] **SO4** Describe vasodilator pharmacology (nitrates, calcium channel blockers) in angina pectoris and its mechanistic overlap with cardiovascular-renal disease management.
- [ ] **SO5** Identify high-risk drug combinations and monitoring needs at the cardiorenal interface, including the "triple whammy," and apply the full renal/cardiorenal toolkit to integrated clinical scenarios with a cumulative recall quiz.